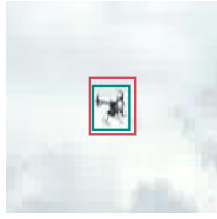
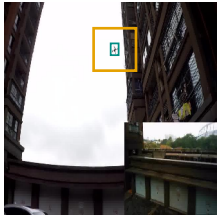
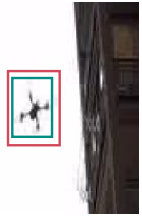
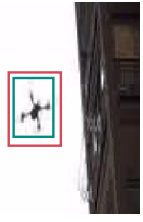
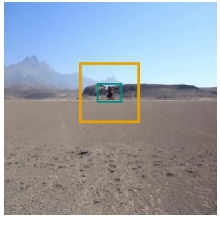
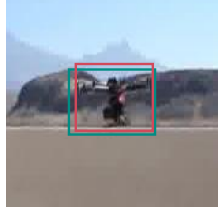
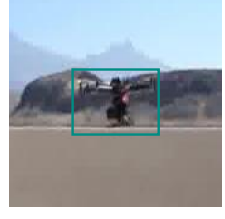
Matched 200-epoch seed-1 checkpoints | original-source test images

GT: teal solid box Prediction: red solid box Crop boundary: amber Display confidence ≥0.25

Scene and crop	YOLO11n	YOLO11n + P2	YOLO11n + P2 + ADown
a Small target / tighter box (24 px)				
		IoU 0.62 | conf 0.60	IoU 0.70 | conf 0.68	IoU 0.78 | conf 0.62
b Small target / tighter box (29 px)				
		IoU 0.61 | conf 0.69	IoU 0.61 | conf 0.57	IoU 0.78 | conf 0.57
c Boundary case / missed target (62 px)				
		IoU 0.78 | conf 0.67	IoU 0.78 | conf 0.55	No IoU≥0.50 detection

Identical native-pixel crop across models within each row; no image enhancement. Examples are illustrative, not a population estimate.